IAE METZ · FINANCE

UNIVERSITÉ DE LORRAINE | IAE METZ School of Management

The Lifeblood of Modern Finance: Financial Data Sources

Terminals · Fundamentals · Credit · APIs · Alternative Data

Sitraka FORLER

“Your models are only as good as the data you feed them.

In the 1980s, a faster phone line gave you an edge.

Today, the edge belongs to those who know WHERE to find the right data and HOW to ingest it efficiently.

THREE MODULES : Terminal Aggregators · Fundamental & Credit Titans · The API Revolution

MODULE 01

The Terminal Aggregators

The Institutional Gold Standard

$30K

per seat / year
(Bloomberg)

330K+

Bloomberg
terminals worldwide

~35%

market share
(LSEG)

Bloomberg Terminal — The Undisputed King

$30,000

per seat
per year

Non-negotiable for real-time market making & macro FX

Real-Time Pricing

Equities, fixed income, FX, and commodities — all asset classes in one unified ecosystem.

IB Chat Network

Bloomberg Instant Messaging — how trading floors talk, trade, and gossip globally.

Fixed Income Data

The deepest bond pricing database in the world. Essential for rates & credit desks.

Bloomberg API (BLPAPI)

Programmatic access to pull real-time & historical data directly into Python or Excel.

LSEG Workspace (formerly Refinitiv Eikon)

	LSEG WORKSPACE	BLOOMBERG TERMINAL
Cost	Lower — more cost-effective	~$30,000/seat/year
Strength	FX via FXall · Fundamental analysts	Real-time trading · IB chat network
Excel Integration	Seamless, native add-in	Available but less fluid
Users	Wealth managers · Fund analysts	Traders · Macro desks · IBD
Historical Data	Deep — excellent coverage	Excellent — slightly less history
Best For	Fundamental analysis · FX	Market making · Rates · Credit

MODULE 02

Fundamental &
Credit Titans

Price action is what happens today — fundamentals dictate what happens tomorrow.

Moody's Analytics

Default Prob · Structured Finance

S&P Global / Capital IQ

Financial Models · Capital Structure

FactSet

Portfolio Analytics · Attribution

CRSP & Compustat

Academic · Survivorship Bias-Free

Moody's Analytics & S&P Global / Capital IQ

MOODY'S ANALYTICS

Credit Ratings Engine

Beyond the headline rating — raw PD (Probability of Default) models used by risk teams at banks and insurers.

Structured Finance Data

CDO, CLO, ABS performance analytics. Critical for structured credit desks.

Corporate Health Scores

Financial statement normalization + early warning scoring for credit deterioration.

KMV / EDF Model

The Expected Default Frequency model — gold standard for quantitative credit risk.

S&P GLOBAL / CAPITAL IQ

Standardized Financials

Meticulously normalized income statements, balance sheets, and cash flows across 90,000+ public companies.

Capital Structure Details

Full debt waterfall, covenant analysis, and equity ownership — staple of LBO & M&A modeling.

Deal & Transaction DB

M&A comps, precedent transactions, and league table data. Essential for IB pitch books.

Screening & Valuation

Comparable company analysis (comps) with pre-built multiples: EV/EBITDA, P/E, EV/Sales.

FactSet · CRSP · Compustat

FACTSET

Beloved by hedge funds & long-only asset managers

- ✓ **Portfolio Attribution**
  See EXACTLY why you under/outperformed the benchmark on any given day. Factor decomposition.
- ✓ **Earnings Estimates**
  Consensus EPS, revenue, and guidance data from the sell-side analyst community.
- ✓ **Portfolio Risk**
  Real-time factor exposure, correlation analysis, and stress testing across asset classes.
- ✓ **Multi-Asset Coverage**
  Equities, fixed income, FX, and alternatives in a single normalized data model.

CRSP — Center for Research in Security Prices

Stock prices since 1926. Survivorship bias-free — critical for long-horizon backtesting. The academic gold standard at Wharton, Chicago, and all top quant programs.

COMPUSTAT — Accounting Data

Standardized financial statements for 100,000+ companies globally. Used alongside CRSP for factor model construction (Fama-French) and long-term quantitative research.

MODULE 03

The API **Revolution**

& Alternative Data

Ten years ago, you had to be a massive institution to access high-quality data.

Today, a developer with a laptop and an API key can compete with tier-1 institutions.

REST APIs · WebSockets · Tick Data · Satellite Imagery · Credit Card Flows · Jet Tracking

API Providers — The Developer-First Ecosystem

Polygon.io

PRO / BOUTIQUE

- Real-time tick data via WebSocket
- REST API — 5+ years of history
- Options chain, crypto, FX
- Sub-second latency for HFT systems

Best for: HFT models · Systematic vol strategies

Alpaca Markets

ALGO TRADING

- Commission-free trading API
- Paper trading environment
- Live brokerage integration
- Event-driven strategy deployment

Best for: Automated execution · Strategy testing

IEX Cloud

FUNDAMENTALS

- Pricing + fundamentals in one API
- Financial statements normalized
- News & sentiment feeds
- CRSP-quality historical pricing

Best for: Fundamental quant models · Screeners

Alpha Vantage

PROTOTYPING

- Free tier — excellent for students
- Daily OHLCV historical data
- Technical indicators pre-built
- Forex & crypto coverage

Best for: Model prototyping · Education · R&D

MODULE 03

Alternative Data

The Frontier of Modern Alpha Generation

Hedge funds are no longer just looking at balance sheets.

🛰️ Satellite Imagery

Walmart parking lot density predicts quarterly same-store sales before the print. Planet Labs, Orbital Insight.

💳 Credit Card Flows

Transaction-level spend data via Yodlee, Earnest Research. Real consumer behavior — no survey lag.

✈ Corporate Jet Tracking

Executives travel before announcements. ADS-B data from FlightAware flags M&A activity in advance.

📱 Mobile Geolocation

Foot traffic to retail stores vs. competitors. SafeGraph, Placer.ai power consumer sector hedge funds.

📋 Web Scraping / NLP

Glassdoor job postings signal capex expansion. Earnings call sentiment NLP detects management stress.

🌡️ Shipping & IoT

Baltic Dry Index, container GPS tracking, and commodity flow sensors. Real economy in real time.

Python — Pull Stock Data via API

LAB

python — yahoo_finance_pull.py

```python
import yfinance as yf
import pandas as pd

# Pull 1 year of AAPL daily data
ticker = yf.Ticker('AAPL')
hist = ticker.history(period='1y')

# Compute log returns
import numpy as np
hist['log_ret'] = np.log(
    hist['Close'] / hist['Close'].shift(1)
)

# Annualized Sharpe Ratio
rf = 0.04 / 252  # daily risk-free
excess = hist['log_ret'] - rf
sharpe = (excess.mean() /
          excess.std()) * np.sqrt(252)

print(f'Sharpe: {sharpe:.3f}')
# Output: Sharpe: 1.241
```

python — polygon_io_pull.py

```python
from polygon import RESTClient
import pandas as pd

# Polygon.io — 1-min OHLCV bars
client = RESTClient('YOUR_API_KEY')

bars = client.get_aggs(
    ticker='AAPL',
    multiplier=1,
    timespan='minute',
    from_='2024-01-01',
    to='2024-01-31'
)

df = pd.DataFrame(bars)
df['timestamp'] = pd.to_datetime(
    df['timestamp'], unit='ms'
)

# 27,000+ 1-min bars in ~0.8s
print(df[['timestamp','close']].head())
```

THE WHARTON TAKEAWAY

The Right Question

is never 'How do I get data?'

The question is: What is the LATENCY, CLEANLINESS, and COST of the data I need to solve this specific financial problem?

⏱️ LATENCY

How fresh does it need to be?

HFT needs microseconds. Daily factor investing is fine with overnight data. Matching latency to strategy is a cost optimization, not a vanity decision.

🧹 CLEANLINESS

Is it survivorship bias-free?

Dirty data builds fake alpha. CRSP scrubs delisted stocks. Raw web-scraped data doesn't. Always audit your universe construction before backtesting.

💰 COST

Does the alpha justify the spend?

Bloomberg at $30K/seat is cheap if you're running $1B. Expensive for a $5M fund. Alternative data at $200K/year only pays if the signal is orthogonal.

Who Uses What — The Practitioner Matrix

IAE METZ

Role	Primary Source	Secondary	Tertiary
Macro FX Trader	Bloomberg (IB chat + FX pricing)	LSEG / FXall	—
Distressed Debt Analyst	Moody's KMV · Capital IQ	Bloomberg (bond pricing)	Compustat (financials)
Long-Only Portfolio Mgr	FactSet (attribution)	Bloomberg / LSEG	Capital IQ (comps)
Quant / Systematic	CRSP + Compustat	Polygon.io / Databento	Alt data (cc flows, geo)
Investment Banker	Capital IQ (comps, deals)	Bloomberg (market data)	Moody's (credit)
Hedge Fund (L/S Equity)	FactSet + Alt Data	Bloomberg	Polygon (tick data)
Academic Researcher	CRSP + Compustat	Alpha Vantage / yfinance	WRDS platform
HFT / Algo Developer	Polygon.io / Databento	IEX Cloud	Co-located feeds

Class Dismissed.

Next Steps for the Hungry !

1. Write a Python script to pull stock data via Polygon.io or yfinance
2. Explore how alternative data generates orthogonal alpha
3. Build your first factor model using CRSP + Compustat via WRDS